Name	
Teacher /Code	
Course/Subject	**NCEA Chemistry & Biology Level 1**
Standard	**AS 92022**

Level 1 Chemistry & Biology 2026

92022 Demonstrate understanding of genetic variation in relation to an identified characteristic

Credits: 5

Achievement	Achievement with Merit	Achievement with Excellence
Demonstrate understanding of genetic variation in relation to an identified characteristic.	Explain genetic variation in relation to an identified characteristic.	Evaluate genetic variation in relation to an identified characteristic
Overall Level of Performance		

You should attempt ALL the questions in this booklet.

For assessor use only:			
Q1	**Q2**	**Q3**	**Total**

Question 1 – Huntington's disease

Huntington's disease is an inherited genetic disorder caused by a dominant mutation in the HTT gene on chromosome 4.

Individuals with the mutant HTT allele often develop symptoms in adulthood, usually after the age of 30. Symptoms may include uncontrollable movements, cognitive decline, and emotional disturbances.

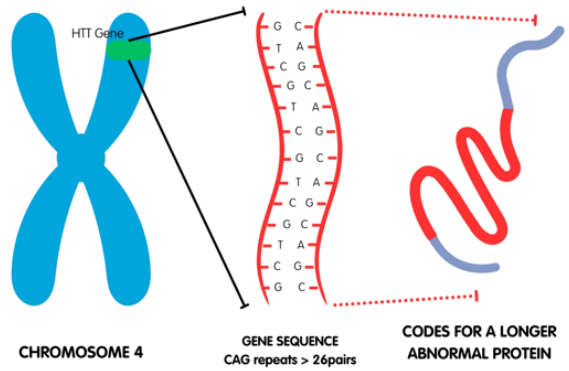


Diagram 1 – Location of the HTT gene on chromosome 4.
https://huntingtonsvic.org.au/huntingtin-gene/

a. Explain the relationship between chromosomes, genes, alleles, and DNA, using Huntington's disease as an example.

b. Explain the difference between somatic and gametic mutations.

c. Discuss why the allele frequency of the mutant HTT allele has not increased in the gene pool, even though the mutant HTT allele can be passed on.
In your answer, provide a reason why the mutant HTT allele can be passed on.

Question 2 – Herbicide resistance

Herbicides are chemicals used to kill unwanted plant species (weeds).

Old Man's Beard (*Clematis vitalba*) is a weed. It is a woody vine plant that is invasive, spreading rapidly and negatively impacting its hosts by smothering them and limiting their ability to photosynthesise. Over many generations, resistant individuals within Old Man's Beard populations have become more common following repeated exposure to herbicides such as glyphosate.

Scientists want to identify the specific gene (or genes) within the plant genome that code for glyphosate resistance.



Diagram 2 – Old Man's Beard (*Clematis vitalba*) growing around cabbage tree (*Cordyline australis*), by Alan, Wikimedia Commons.

a. Discuss how sexual reproduction produces genetic variation in the Old Man's Beard populations.
In your answer:
- describe sexual reproduction and fertilisation,
- explain how meiosis produces genetically different gametes
- explain how this leads to variation in the Old Man's Beard population.

b. Discuss how the frequency of the resistant allele increases in the Old Man's Beard population.
In your answer:

- describe mutation, genome, and allele frequency,
- explain why there are more weed plants that are resistant to herbicide.

Question 3 – Yellow-eyed penguin

The yellow eyed penguin (*Megadyptes antipodes*), or hoiho, is one of the world's rarest birds. Hoiho populations in both the North and South Island in New Zealand have severely declined over the past two decades.

On the Otago Peninsula (South Island), the hoiho populations have decreased by 75% since the mid-1990s. Current trends predict the population could disappear from the area entirely in the next 40 years.

The Yellow-eyed Penguin Trust is a conservation program that has set up a protected area with more than 48 pairs of breeding hoiho. This represents more than 10% of the total hoiho population across all of New Zealand and the sub-Antarctic islands.

Scientists often use gene tracking methodologies, such as DNA sequencing, DNA profiling, and pedigree analysis, as part of the hoiho conservation programme.



Diagram 3 – Hoiho on a rocky shore by Christian, Wikimedia Commons.

a. Describe one purpose of a gene tracking methodology.

b. Discuss how scientists could use one gene tracking methodology in the hoiho conservation programme.

c. Evaluate the effectiveness of the hoiho conservation program

In your answer, include:

- a description of genetic variation, genotype and phenotype,
- an explanation of why genetic variation is important for the hoiho population,
- evaluate whether the conservation programme is likely to successful, using biological evidence.

Extra paper/space if required. Write the question number(s) clearly.